

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name Marshal® 35 DS insecticide

Other means of identification

Product code 50000368

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-stance/Mixture : Can be used as insecticide only.

Recommended restrictions on use : Use as recommended by the label.

1.3 Details of the supplier of the safety data sheet

Supplier Address

FMC Agricultural Solutions A/S
Thyborønvej 78
DK-7673 Harboøre
Denmark

Telephone: +45 9690 9690
Telefax: +45 9690 9691
E-mail address: SDS-Info@fmc.com .

1.4 Emergency telephone

For leak, fire, spill or accident emergencies, call:
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical emergency:
In case of poisoning call toll free number: 0800720021 or
0800730030 (24 hrs/masaa 24)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Acute toxicity, Category 3 H301: Toxic if swallowed.

Acute toxicity, Category 2 H330: Fatal if inhaled.

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Specific target organ toxicity - single exposure, Category 1

H370: Causes damage to organs.

Specific target organ toxicity - repeated exposure, Category 1

H372: Causes damage to organs through prolonged or repeated exposure.

Short-term (acute) aquatic hazard, Category 1

H400: Very toxic to aquatic life.

Long-term (chronic) aquatic hazard, Category 1

H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labeling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal Word :

DANGER

Hazard Statements :

H301 Toxic if swallowed.
H330 Fatal if inhaled.
H370 Causes damage to organs.
H372 Causes damage to organs through prolonged or repeated exposure.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P260 Do not breathe dust.
P273 Avoid release to the environment.

Response:

P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. Rinse mouth.
P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.
P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

Hazardous ingredients which must be listed on the label:

carbosulfan (ISO)
2,2' -oxybisethanol

Additional Labeling

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

EUH208 Contains carbosulfan (ISO). May produce an allergic reaction.

EUH401 To avoid risks to human health and the environment, comply with the instructions for use.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Dust can form an explosive mixture in air.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
carbosulfan (ISO)	55285-14-8 259-565-9 006-084-00-5	Acute Tox. 3; H301 Acute Tox. 2; H330 Skin Sens. 1; H317 STOT SE 1; H370 (Nervous system, Bladder, Gastro- intestinal system, Blood) STOT RE 1; H372 (Nervous system, Bladder, Gastro- intestinal system, Blood) Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 100 M-Factor (Chronic aquatic toxicity): 10	>= 30 - < 50
2,2' -oxybisethanol	111-46-6 203-872-2 603-140-00-6	Acute Tox. 4; H302	>= 1 - < 10
9-(2-carboxyphenyl)-3,6-	81-88-9	Skin Irrit. 2; H315	>= 1 - < 2.5

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

bis(diethylamino)xanthylum chloride	201-383-9	Eye Dam. 1; H318 STOT SE 3; H335 (Respiratory system) Aquatic Chronic 3; H412
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For explanation of abbreviations see section 16.

SECTION 4: First aid measures
4.1 Description of first-aid measures

- | | | |
|----------------------------|---|--|
| General advice | : | Move out of dangerous area.
Consult a physician.
Show this material safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended. |
| Protection of first-aiders | : | Avoid inhalation, ingestion and contact with skin and eyes. |
| If inhaled | : | Call a physician or poison control center immediately.
If unconscious, place in recovery position and seek medical advice. |
| In case of skin contact | : | If on skin, rinse well with water.
If on clothes, remove clothes.
Wash off with soap and plenty of water.
Get medical attention immediately if irritation develops and persists.
Wash contaminated clothing before re-use. |
| In case of eye contact | : | Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist. |
| If swallowed | : | Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
Take victim immediately to hospital. |

4.2 Most important symptoms and effects, both acute and delayed

- | | | |
|-------|---|---|
| Risks | : | Toxic if swallowed.
Fatal if inhaled.
Causes damage to organs.
Causes damage to organs through prolonged or repeated exposure. |
|-------|---|---|

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.
It may be helpful to show this safety data sheet to physician.
This product contains a reversible cholinesterase inhibitor.
Atropine sulfate is antidotal. If cyanosis is absent: Adults - start treatment by giving 2 mg atropine intravenously or intramuscularly, if necessary, and repeat with 0.4 - 2.0 mg atropine at 15 minute intervals until atropinization occurs (tachycardia, flushed skin, dry mouth, mydriasis); Children under 12 - initial dose = 0.05 mg/kg body weight and repeat dose = 0.02 - 0.05 mg/kg body weight. Use of oximes such as 2-PAM is controversial. Observe patient to ensure that these symptoms do not recur as atropinization wears off. If in eyes, instill one drop of homatropine. Treatment is otherwise controlled removal of exposure followed by symptomatic and supportive care. Immediate medical attention is required in case of ingestion.

SECTION 5: Firefighting measures**5.1 Extinguishing media**

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet
Do not spread spilled material with high-pressure water streams.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Carbon oxides
Sulfur oxides
Nitrogen oxides (NO_x)
Hydrogen cyanide

5.3 Advice for firefighters

Special protective equipment for fire-fighters : Wear self-contained breathing apparatus for firefighting if necessary.

Further information : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

SECTION 6: Accidental release measures**6.1 Personal precautions, protective equipment and emergency procedures**

Personal precautions : Use personal protective equipment.
Avoid dust formation.
Avoid breathing dust.
Ensure adequate ventilation.
Evacuate personnel to safe areas.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.

6.2 Environmental precautions

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage**7.1 Precautions for safe handling**

Advice on safe handling : Avoid formation of respirable particles.
Do not breathe vapors/dust.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Advice on protection against fire and explosion : Avoid dust formation. Provide appropriate exhaust ventilation at places where dust is formed.

Hygiene measures : Avoid contact with skin, eyes and clothing. When using do not eat or drink. When using do not smoke. Wash hands before breaks and immediately after handling the product. Remove and wash contaminated clothing and gloves, including the inside, before re-use.

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000368 Date of last issue: -
Date of first issue: 24.09.2025

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Prevent unauthorized access. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards.

Further information on storage conditions : Store in closed, labelled containers. The storage room should be constructed of incombustible material, closed, dry, ventilated and with impermeable floor, without access of unauthorised persons or children. A warning sign reading "POISON" is recommended. The room should only be used for storage of chemicals. Food, drink, feed and seed should not be present. A hand wash station should be available.

Further information on storage stability : No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s) : Registered pesticide to be used in accordance with a label approved by country-specific regulatory authorities.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
Silicic acid, calcium salt	1344-95-2	TWA (Respirable dust)	5 mg/m ³	KE OEL
Further information: Recommended Limit				
		TWA (inhalable dust)	10 mg/m ³	KE OEL
Further information: Recommended Limit				
2,2' -oxybisethanol	111-46-6	TWA	23 ppm 100 mg/m ³	KE OEL
Further information: Recommended Limit				

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Routes of exposure	Potential health effects	Value
Poly(oxy-1,2-ethanediyl),.alpha.-hydro-.omega.-hydroxy-	Workers	Inhalation	Long-term systemic effects	40.2 mg/m ³
	Workers	Dermal	Long-term systemic effects	112 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	7.14 mg/m ³

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000368 Date of last issue: -
Date of first issue: 24.09.2025

	Consumers	Oral	Long-term systemic effects	40 mg/kg bw/day
Soybean oil, epoxidized	Workers	Inhalation	Long-term systemic effects	11.9 mg/m3
	Workers	Inhalation	Acute systemic effects	70 mg/m3
	Workers	Dermal	Long-term systemic effects	1.7 mg/kg bw/day
	Workers	Dermal	Acute systemic effects	10 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	2.8 mg/m3
	Consumers	Inhalation	Acute systemic effects	17.5 mg/m3
	Consumers	Dermal	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Dermal	Acute systemic effects	5 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Oral	Acute systemic effects	5 mg/kg bw/day
2,2' -oxybisethanol	Workers	Inhalation	Long-term local effects	60 mg/m3
	Workers	Dermal	Long-term systemic effects	43 mg/kg
	Consumers	Inhalation	Long-term systemic effects	12 mg/m3
	Consumers	Inhalation	Long-term local effects	12 mg/m3
	Consumers	Dermal	Long-term systemic effects	21 mg/kg

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
Silicic acid, calcium salt	Fresh water	4 mg/l
	Sea water	
	Fresh water sediment	
	Sea sediment	
	Soil	
Poly(oxy-1,2-ethanediyl),.alpha.-hydro-.omega.-hydroxy-	Fresh water	273 mg/l
	Intermittent use/release	1 mg/l
	Sea water	27.3 mg/l
	Intermittent use/release	0.1 mg/l
	Fresh water sediment	1030 mg/kg dry weight (d.w.)
	Sea sediment	103 mg/kg dry weight (d.w.)
	Soil	46.4 mg/kg dry weight (d.w.)
Soybean oil, epoxidized	Soil	6.25 mg/kg dry weight (d.w.)

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000368 Date of last issue: -
Date of first issue: 24.09.2025

2,2' -oxybisethanol	Fresh water	10 mg/l
	Sea water	1 mg/l
	Intermittent use/release	10 mg/l
	Sewage treatment plant	199.5 mg/l
	Fresh water sediment	20.9 mg/kg dry weight (d.w.)
	Sea sediment	2.09 mg/kg dry weight (d.w.)
	Soil	1.53 mg/kg dry weight (d.w.)

8.2 Exposure controls

Personal protective equipment

- Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles
- Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
- Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.
- Skin and body protection : Dust impervious protective suit
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
- Respiratory protection : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.
- Filter type : Particulates type (P)
- Protective measures : Plan first aid action before beginning work with this product.
Always have on hand a first-aid kit, together with proper instructions.
Wear suitable protective equipment.
When using do not eat, drink or smoke.
In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

- Physical state : solid
Form : Dry powder
Color : red
Odor : phenol-like

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

pH	:	No data available
Melting point/freezing point	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	Not applicable
Flammability (solid, gas)	:	Not classified as a flammability hazard
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Density	:	16 - 20 lb/scf
Solubility(ies)	:	
Water solubility	:	No data available
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	Not applicable
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity	:	
Viscosity, kinematic	:	Not applicable
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing

9.2 Other information

Particle size	:	No data available
Particle Size Distribution	:	No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No decomposition if stored and applied as directed.

10.2 Chemical stability

No decomposition if stored and applied as directed.

10.3 Possibility of hazardous reactions

Hazardous reactions	:	Contact with aqueous acids may produce carbofuran.
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No decomposition if stored and applied as directed.

10.4 Conditions to avoid

Conditions to avoid	:	Heat, flames and sparks.
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10.5 Incompatible materials

Materials to avoid	:	Avoid strong acids, bases, and oxidizers.
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Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

10.6 Hazardous decomposition products

No hazardous decomposition products are known.
See subsection 5.2.

SECTION 11: Toxicological information**11.1 Information on toxicological effects****Acute toxicity**

Toxic if swallowed.
Fatal if inhaled.

Product:

Acute oral toxicity	:	LD50 (Rat, male and female): 131 mg/kg Method: OECD Test Guideline 401
Acute inhalation toxicity	:	LC50 (Rat, male and female): 0.14 mg/l Exposure time: 4 h Test atmosphere: dust/mist Remarks: Based on data from a similar product.
Acute dermal toxicity	:	LD50 (Rat, male and female): > 2,000 mg/kg Method: OECD Test Guideline 402 Assessment: The substance or mixture has no acute dermal toxicity

Components:**carbosulfan (ISO):**

Acute oral toxicity	:	LD50 (Rat, female): 185 mg/kg
Acute inhalation toxicity	:	LC50 (Rat, female): 0.15 mg/l Exposure time: 4 h Test atmosphere: dust/mist
Acute dermal toxicity	:	LD50 (Rat): > 2,000 mg/kg

2,2' -oxybisethanol:

Acute oral toxicity	:	LD50 (Rat, male and female): 16,500 mg/kg
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9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylium chloride:

Acute oral toxicity	:	LD50 (Rat): > 2,000 mg/kg
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Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Components:**carbosulfan (ISO):**

Species	: Rabbit
Result	: slight irritation

2,2' -oxybisethanol:

Species	: reconstructed human epidermis (RhE)
Method	: OECD Test Guideline 439
Result	: No skin irritation

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Species	: Rabbit
Result	: Skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	: Rabbit
Method	: OECD Test Guideline 405
Result	: No eye irritation
Remarks	: Minimal effects that do not meet the threshold for classification.

Components:**carbosulfan (ISO):**

Species	: Rabbit
Result	: slight irritation

2,2' -oxybisethanol:

Species	: Rabbit
Result	: No eye irritation

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Result	: Irreversible effects on the eye
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Respiratory or skin sensitization**Skin sensitization**

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Routes of exposure	: Skin contact
Species	: Guinea pig

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Method	:	US EPA Test Guideline OPP 81-6
Result	:	Does not cause skin sensitization.
Remarks	:	Based on data from a similar product.

Components:

carbosulfan (ISO):

Test Type	:	Buehler Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.
Test Type	:	Patch test
Species	:	Guinea pig
Result	:	May cause sensitization by skin contact.

2,2' -oxybisethanol:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	Regulation (EC) No. 440/2008, Annex, B.6
Result	:	Does not cause skin sensitization.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

carbosulfan (ISO):

Genotoxicity in vitro	:	Test Type: reverse mutation assay Test system: Salmonella typhimurium Result: negative Test Type: reverse mutation assay Test system: Escherichia coli Result: negative Test Type: gene mutation test Test system: Chinese hamster cells Result: negative Test Type: Chromosome aberration test in vitro Test system: Chinese hamster cells Result: negative
Genotoxicity in vivo	:	Test Type: chromosome aberration assay Species: mice Result: negative

2,2' -oxybisethanol:

Genotoxicity in vitro	:	Test Type: reverse mutation assay Method: OECD Test Guideline 471
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Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male)
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Genotoxicity in vitro : Test Type: reverse mutation assay
Test system: Salmonella typhimurium
Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Result: negative

Test Type: gene mutation test
Test system: Chinese hamster lung cells
Method: OECD Test Guideline 476
Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:**carbosulfan (ISO):**

Species : Mouse
Exposure time : 2 Years
NOAEL : 2.5 mg/kg bw/day
Result : negative

Species : Rat
Exposure time : 2 Years
NOAEL : 1 mg/kg bw/day
Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

2,2' -oxybisethanol:

Species : Rat, male and female
Application Route : Oral
Exposure time : 108 weeks
Result : negative

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Species : Mouse, female
Application Route : Dermal
Exposure time : 18 month(s)
Dose : 121.4 mg
Result : negative

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:**carbosulfan (ISO):**

Effects on fertility : Test Type: Three-generation study
Species: Rat
Application Route: Oral
General Toxicity Parent: NOAEL: 1.2 mg/kg bw/day
Fertility: NOAEL: 1.2 mg/kg bw/day
Result: negative

Effects on fetal development : Test Type: Embryo-fetal development
Species: Rat
Application Route: Oral
General Toxicity Maternal: NOAEL: 2 mg/kg bw/day
Developmental Toxicity: NOAEL: 2
Result: negative

Test Type: Embryo-fetal development
Species: Rabbit
Application Route: Oral
General Toxicity Maternal: NOAEL: 5 mg/kg bw/day
Developmental Toxicity: NOAEL: 10
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

2,2' -oxybisethanol:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Mouse, male and female
Application Route: Oral
Result: negative

Effects on fetal development : Species: Rat
Application Route: Oral
Method: OECD Test Guideline 414
Result: Embryotoxic effects and adverse effects on the offspring were detected only at high maternally toxic doses

STOT-single exposure

Causes damage to organs.

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Components:**carbosulfan (ISO):**

Target Organs	: Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment	: The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Assessment	: May cause respiratory irritation.
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STOT-repeated exposure

Causes damage to organs through prolonged or repeated exposure.

Components:**carbosulfan (ISO):**

Target Organs	: Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment	: The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Assessment	: The substance or mixture is not classified as specific target organ toxicant, repeated exposure.
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Repeated dose toxicity**Components:****carbosulfan (ISO):**

Species	: Rat
NOAEL	: 2 mg/kg bw/day
Application Route	: Oral
Exposure time	: 90 days

Species	: Dog
NOAEL	: 1.6 mg/kg bw/day
Application Route	: Oral
Exposure time	: 6 months

2,2' -oxybisethanol:

Species	: Rat, male and female
NOAEL	: 300 mg/kg
Application Route	: Oral
Exposure time	: 98 d

Species	: Dog, male
NOAEL	: 2,220 mg/kg
Exposure time	: 28 d
Method	: OECD Test Guideline 410

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylium chloride:

Species	: Rat
Application Route	: Oral
Exposure time	: 12 weeks
Dose	: 50400 mg/kg/12W-C
Symptoms	: Reduced body weight

Species	: Rat
Application Route	: Inhalation
Exposure time	: 13 weeks
Dose	: 110 mg/kg/13W-I
Symptoms	: Gained body weight

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

carbosulfan (ISO):

The substance does not have properties associated with aspiration hazard potential.

Neurological effects

Components:

carbosulfan (ISO):

Remarks	: Neurotoxicity observed in animals studies
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Further information

Product:

Remarks	: No data available
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SECTION 12: Ecological information

12.1 Toxicity

Components:

carbosulfan (ISO):

Toxicity to fish	: LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.015 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	: EC50 (Daphnia magna (Water flea)): 0.0015 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	: EC50 (Pseudokirchneriella subcapitata (microalgae)): > 20 mg/l Exposure time: 96 h

M-Factor (Acute aquatic tox-	: 100
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Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

icity)

Toxicity to fish (Chronic toxicity) : NOEC: 0.00828 mg/l
Exposure time: 21 d
Species: Pimephales promelas (fathead minnow)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0.0032 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to terrestrial organisms : 1.035 µg/bee
Species: Apis mellifera (bees)
Remarks: Oral

0.18 µg/bee
Species: Apis mellifera (bees)
Remarks: Contact

LD50: 10 mg/kg
Species: Anas platyrhynchos (Mallard duck)

2,2' -oxybisethanol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 75,200 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 10,000 mg/l
Exposure time: 24 h

Toxicity to algae/aquatic plants : NOEC (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l
Method: OECD Test Guideline 201

Toxicity to microorganisms : EC20 (activated sludge): > 1,995 mg/l
Exposure time: 30 min
Test Type: Respiration inhibition
Method: ISO 8192

Toxicity to fish (Chronic toxicity) : NOEC: 15,380 mg/l
Exposure time: 7 d
Species: Pimephales promelas (fathead minnow)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 8,590 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia (water flea)

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Toxicity to fish : LC50 (Oryzias latipes (Orange-red killifish)): > 12 mg/l
Exposure time: 48 h

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

LC50 (Lepomis macrochirus (Bluegill sunfish)): 100 mg/l
Exposure time: 96 h

LC50 (Oncorhynchus mykiss (rainbow trout)): 217 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 22.9 mg/l
Exposure time: 48 h

12.2 Persistence and degradability**Components:****carbosulfan (ISO):**

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 28 %
Exposure time: 28 d

Stability in water : Remarks: Hydrolyzes readily.

2,2' -oxybisethanol:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 25 - 92 %
Exposure time: 28 d

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylum chloride:

Biodegradability : Inoculum: activated sludge
Result: Not readily biodegradable.
Biodegradation: 0 %
Exposure time: 28 d

12.3 Bioaccumulative potential**Components:****carbosulfan (ISO):**

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 990
Remarks: Can accumulate in aquatic organisms.

Partition coefficient: n-octanol/water : log Pow: 5.37
pH: 8
Method: OECD Test Guideline 107

2,2' -oxybisethanol:

Bioaccumulation : Species: Leuciscus idus (Golden orfe)
Bioconcentration factor (BCF): 100

Partition coefficient: n-octanol/water : log Pow: -1.98 (20 °C)

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

9-(2-carboxyphenyl)-3,6-bis(diethylamino)xanthylium chloride:

Partition coefficient: n-octanol/water : log Pow: 2

12.4 Mobility in soil**Components:****carbosulfan (ISO):**

Distribution among environmental compartments : Remarks: Slightly mobile in soils

Stability in soil :

12.5 Results of PBT and vPvB assessment**Product:**

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects**Product:**

Endocrine disrupting potential : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

SECTION 13: Disposal considerations**13.1 Waste treatment methods**

Product : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Triple rinse containers.
Do not re-use empty containers.
Packaging that is not properly emptied must be disposed of as the unused product.

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number

UNRTDG	:	UN 2757
IMDG	:	UN 2757
IATA	:	UN 2757

14.2 UN proper shipping name

UNRTDG	:	CARBAMATE PESTICIDE, SOLID, TOXIC (Carbosulfan)
IMDG	:	CARBAMATE PESTICIDE, SOLID, TOXIC (Carbosulfan)
IATA	:	Carbamate pesticide, solid, toxic (Carbosulfan)

14.3 Transport hazard class(es)

	Class	Subsidiary risks
UNRTDG	:	6.1
IMDG	:	6.1
IATA	:	6.1

14.4 Packing group

UNRTDG		
Packing group	:	II
Labels	:	6.1
IMDG		
Packing group	:	II
Labels	:	6.1
EmS Code	:	F-A, S-A
IATA (Cargo)		
Packing instruction (cargo aircraft)	:	676
Packing instruction (LQ)	:	Y644
Packing group	:	II
Labels	:	Toxic
IATA (Passenger)		
Packing instruction (passenger aircraft)	:	669
Packing instruction (LQ)	:	Y644
Packing group	:	II
Labels	:	Toxic

14.5 Environmental hazards

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

UNRTDG

Environmentally hazardous : yes

IMDG

Marine pollutant : yes

IATA (Passenger)

Environmentally hazardous : yes

IATA (Cargo)

Environmentally hazardous : yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture****The ingredients of this product are reported in the following inventories:**

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. carbosulfan (ISO)
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: On the inventory, or in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

15.2 Chemical Safety Assessment

A chemical safety assessment is not required for this product (mixture).

SECTION 16: Other information
Full text of H-Statements

H301	: Toxic if swallowed.
H302	: Harmful if swallowed.
H315	: Causes skin irritation.
H317	: May cause an allergic skin reaction.
H318	: Causes serious eye damage.
H330	: Fatal if inhaled.
H335	: May cause respiratory irritation.
H370	: Causes damage to organs.
H372	: Causes damage to organs through prolonged or repeated exposure.
H400	: Very toxic to aquatic life.
H410	: Very toxic to aquatic life with long lasting effects.
H412	: Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Acute	: Short-term (acute) aquatic hazard
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Skin Irrit.	: Skin irritation
Skin Sens.	: Skin sensitization
STOT RE	: Specific target organ toxicity - repeated exposure
STOT SE	: Specific target organ toxicity - single exposure
KE OEL	: Kenya. Occupational Exposure Limits
KE OEL / TWA	: TWA OEL

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECl - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office

SAFETY DATA SHEET



Marshal® 35 DS insecticide

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000368	Date of first issue: 24.09.2025

of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information :

Classification of the mixture:

Acute Tox. 3	H301
Acute Tox. 2	H330
STOT SE 1	H370
STOT RE 1	H372
Aquatic Acute 1	H400
Aquatic Chronic 1	H410

Classification procedure:

Based on product data or assessment
Based on product data or assessment
Calculation method
Calculation method
Calculation method
Calculation method

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